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BACHELOR'S THESIS

**Integration in Finite Terms:
Liouville's Theorem.**

*Integrazione in termini finiti:
Il Teorema di Liouville.*

(Italian original title)

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1 Introduction

Every student of Calculus I who tries to integrate functions such as e^{x^2} , $\sin(x^2)$, $e^{1/x}$ is told that they are not elementarily integrable, that is, it is not possible to write their antiderivative by combining in a finite way the four elementary operations, raising to powers and extracting roots, logarithms, exponentials, trigonometric and inverse trigonometric functions. But why? Let us give another example of an integrable function that does not admit an elementary antiderivative. In number theory there is much interest in the study of the function $\pi(x) = |\{1 \leq p \leq x : p \text{ prime}\}|$ of the real variable x , which counts the number of primes up to x . Defining the logarithmic integral function $\text{Li}(x) = \int_2^x \frac{dt}{\ln(t)}$, it has been proved that, as $x \rightarrow \infty$, $\pi(x) \sim \text{Li}(x)$. This example shows how important the computation of $\text{Li}(x)$ can be, which however is not expressible in terms of elementary functions. How is it possible to prove a result like this? Certainly, first of all it is necessary to give a precise definition of an elementary function, and the tool that will allow us to move within this domain with ease is differential algebra. Thanks to it we will prove results (first and foremost Liouville's theorem) with which we will give a characterisation of the elementarily integrable functions.

2 Prerequisites

The notions reported in this section are taken from [2].

A *field* F is a commutative ring with unit in which every nonzero element is invertible; we denote by $F^* = F \setminus \{0\}$. Let F, E be fields. E is said to be an *extension* of F (and one writes $E \mid F$) if there exists an injective field homomorphism (called an *embedding*):

$$\sigma : F \longrightarrow E.$$

Let F be a field; we define by $F[x]$ the ring of polynomials with coefficients in F and by $F(x)$ the field of fractions of $F[x]$. Thus

$$F(x) = \left\{ \frac{f}{g} : f, g \in F[x], g \neq 0 \right\}.$$

Note that $F(x)$ is a field and $F(x) \mid F$ is a field extension.

Definition. Let $E \mid F$ be a field extension. Let $b \in E$. We say that b is **algebraic** over F if there exists a polynomial $f \neq 0$ in $F[x]$ such that $f(b) = 0$. An extension $E \mid F$ is said to be an **algebraic extension** if every element of E is algebraic over F .

Definition. If $E \mid F$ is a field extension, then it is possible to regard E in a natural way as a vector space over F . The dimension of E as a vector space over F is called the **degree** of E over F and is denoted by $[E : F]$. If $[E : F] < \infty$, then $E \mid F$ is said to be a **finite** extension.

We also observe that every algebraic extension is finite.

Definition. A field extension $E \mid F$ is said to be **normal** if every element $b \in E$ is algebraic over F and its minimal polynomial factors completely in $E[x]$.

Definition. A polynomial f is said to be *separable* if each of its irreducible factors has all its roots simple (in an extension containing all its roots). A field extension $E \mid F$ is said to be **separable** (over F) if it is algebraic and for every $b \in E$, the minimal polynomial $\min_F(b)$ is separable over F .

Definition. Let R be a ring and let $1 \in R$ be its multiplicative unit. The *characteristic* of R is 0 if $n \cdot 1 \neq m \cdot 1$ for all $n, m \in \mathbb{Z}$; otherwise it is the smallest natural number $n > 0$ such that $n \cdot 1 = 0$.

Proposition. Let F be a field of characteristic 0, or a finite field. Then every extension $E \mid F$ is separable.

Every field that we consider from now on will have characteristic 0 (as for example $\mathbb{Q}$, $\mathbb{R}$ or $\mathbb{C}$).

Definition. A field extension $E | F$ which is finite, normal and separable is said to be a **Galois** extension.

Let us give some reminders about Galois theory, which will be used in what follows.

Definition. Let $E | F$ be a field extension. The set of the F -automorphisms of E (that is, of the automorphisms of E that fix every element of F) is a subgroup of the group $\text{Aut}(E)$ and is denoted by $\text{Gal}(E | F)$. Moreover, given $f \in F[x]$, one defines the *Galois group* of f over F as $\text{Gal}(E | F)$ with E , where E is the splitting field of f over F . Now let H be a subgroup of $\text{Gal}(E | F)$; we define the *field of invariants* $\text{Inv}_E(H) = \{b \in E \mid \sigma(b) = b \ \forall \sigma \in H\}$.

For the proof of the following two propositions see [2].

Proposition 1. *Let E be the splitting field of a separable polynomial $f \in F[x]$ of degree n . Then the subgroup of S_n isomorphic to the Galois group of f over F is transitive if and only if f is irreducible over F .*

Proposition 2. *Let $E | F$ be a Galois extension; then $\text{Inv}_E(E) = F$.*

Symmetric polynomials in $F[x_1, \dots, x_n]$

Definition. Let F be a field, $f \in F[x_1, \dots, x_n]$. We say that f is *symmetric* if for every permutation $\sigma \in S_n$ (the group of permutations of n elements) one has that $f(x_{\sigma(1)}, \dots, x_{\sigma(n)}) = f(x_1, \dots, x_n)$.

Theorem 1. *Let F be a field, $f \in F[x]$ a monic polynomial of degree $n > 0$ with roots $a_1, \dots, a_n$ in one of its splitting fields over F . Then for every symmetric polynomial $p \in F[x_1, \dots, x_n]$ one has that $p(a_1, \dots, a_n) \in F$.*

Proof. Let f be a polynomial as in the hypotheses and let E be its splitting field; then $E | F$ is a Galois extension. We define $G := \text{Gal}(E | F)$ and we consider $p \in F[x_1, \dots, x_n]$ a symmetric polynomial. Every element permutes the roots of f , hence for every $\sigma \in \text{Gal}(E | F)$ one has that $\sigma(p(a_1, \dots, a_n)) = p(a_{\sigma(1)}, \dots, a_{\sigma(n)}) = p(a_1, \dots, a_n)$, and therefore, by Proposition 2, it follows that $p \in \text{Inv}_G(G) = F$. $\square$

Partial fraction decomposition

A standard procedure for the computation of the integrals of rational functions of the type

$$\int \frac{dx}{x^2 - 1}$$

is to break up the integrand (which we do not know how to integrate) into a sum of other, simpler rational functions (which we do know how to integrate). For example, in our case

$$\int \frac{dx}{x^2 - 1} = \int \frac{1}{2} \left(\frac{1}{x - 1} - \frac{1}{x + 1} \right) dx = \int \frac{1}{2(x - 1)} dx - \int \frac{1}{2(x + 1)} dx.$$

But is it always possible to act in this way no matter which rational function is chosen, that is, for any polynomials P, Q in $F[x]$?

Let us begin by fixing a field F and choosing a rational function $v = \frac{P}{Q}$ with $P, Q \in F[x]$ coprime. We factor Q into irreducible factors:

$$Q = Q_1^{k_1} \cdot Q_2^{k_2} \cdots Q_N^{k_N}.$$

Note in particular that $Q_1^{k_1}$ and $Q_2^{k_2} \cdots Q_N^{k_N}$ are coprime, and therefore by Bézout's identity we know that there exist two polynomials T and S such that $TQ_1^{k_1} + SQ_2^{k_2} \cdots Q_N^{k_N} = 1$, and hence we can write

$$v = \frac{P}{Q} = \frac{P(TQ_1^{k_1} + SQ_2^{k_2} \cdots Q_N^{k_N})}{Q_1^{k_1} \cdot Q_2^{k_2} \cdots Q_N^{k_N}} = \frac{PT}{Q_2^{k_2} \cdots Q_N^{k_N}} + \frac{PS}{Q_1^{k_1}}.$$

What have we gained? We have obtained two rational functions in which one denominator is a power of an irreducible polynomial, while the other denominator has one factor fewer than Q in its factorisation. Applying this procedure $N - 1$ times we obtain an expression of the type

$$v = \frac{T_1}{Q_1^{k_1}} + \frac{T_2}{Q_2^{k_2}} + \cdots + \frac{T_N}{Q_N^{k_N}} = \sum_{j=1}^N \frac{T_j}{Q_j^{k_j}}$$

where each Q_j is irreducible. If now the degree of T_j were greater than or equal to the degree of Q_j , we could use Euclid's algorithm to write $T_j = U_j Q_j + R_j$ where R_j is a polynomial of degree less than that of Q_j . As a result we obtain

$$\frac{T_j}{Q_j^{k_j}} = \frac{U_j}{Q_j^{k_j-1}} + \frac{R_j}{Q_j^{k_j}}.$$

In a finite number of steps we arrive at an identity of the form

$$\frac{R_j}{Q_j^{k_j}} = S_{0,j} + \sum_{l=1}^{m_j} \frac{S_{l,j}}{Q_j^l}$$

where, for $l \geq 1$, each polynomial $S_{l,j}$ has degree strictly less than Q_j . We have thus proved the following theorem:

Theorem 2. *Let F be a field, $v \in F(x)$ and $P, Q \in F[x]$ such that $v = \frac{P}{Q}$. Let $Q = Q_1^{k_1} \cdots Q_N^{k_N}$ be its factorisation into irreducibles. Then there exist polynomials $T, S_{ij} \in F[x]$ such that the degree of $S_{i,j}$ is strictly less than the degree of Q_j for every $i \geq 1$, and moreover*

$$v = T + \sum_{i=1}^n \sum_{j=1}^{k_i} \frac{S_{ij}}{Q_i^j}.$$

Such an expression of v is called the partial fraction decomposition.

3 Differential Fields

Definitions and properties of differential fields

Definition 1. A **differential ring** is a pair $(A, ')$ where A is a ring and $' : A \rightarrow A$ a map (called a *derivation*) such that:

$$(D1) \quad (x + y)' = x' + y' \text{ for all } a, b \in A;$$

$$(D2) \quad (xy)' = x'y + xy' \text{ for all } a, b \in A.$$

If A is a field, we will say that $(A, ')$ is a **differential field**.

In general a ring can have more than one derivation defined on it. For example, the ring of polynomials with rational coefficients in two indeterminates $\mathbb{Q}[x, y]$ has at least three derivations: 0 , d/dx , d/dy ; but in fact it has many more: every linear combination of derivations with coefficients in the ring is a derivation on the ring. An example of a differential field, which is then the one that will accompany us along this path, is $(\mathbb{C}(x), ')$ where $'$ is the usual derivative. Indeed $\mathbb{C}(x)$ is a field and $'$ a map that satisfies properties (D1) and (D2).

Proposition 3. Let $(F, ')$ be a differential field; then

1. $1'_F = 0'_F = 0_F$;
2. $(x^k)' = kx^{k-1}x'$ for every $x \in F$ and every $k \in \mathbb{Z}$;
3. $(-x)' = -x'$ for every $x \in F$;
4. $\left(\frac{x}{y}\right)' = \frac{x'y - xy'}{y^2}$ for every $x, y \in F, y \neq 0$;
5. $\frac{(x_1^{k_1}x_2^{k_2}\cdots x_n^{k_n})'}{x_1^{k_1}x_2^{k_2}\cdots x_n^{k_n}} = k_1\frac{x'_1}{x_1} + k_2\frac{x'_2}{x_2} + \cdots k_n\frac{x'_n}{x_n}$ for every $x_1, \dots, x_n \in F^*$ and $k_1, \dots, k_n \in \mathbb{Z}$.

A (rather routine) proof of these properties can be found in [4].

Definition 2. We call the **subfield of constants** of $(F, ')$ the set

$$C_F = \{x \in F : x' = 0\}.$$

Observation. It is not difficult to prove that C_F is a field.

Extensions of differential fields

Definition 3. Let $(F, ')$ and $(E, *)$ be two differential fields. We say that $(E, *)$ is a **differential extension** of $(F, ')$ if the following properties hold:

- (i) E is an extension of F . In symbols $E \mid F$;
- (ii) $*$ extends $'$. In symbols $x^* = x'$ for every $x \in F$.

Let $(F, ')$ be a differential field. We consider the polynomial ring $F[x]$ and the maps $D_0, D_1 : F[x] \rightarrow F[x]$ defined as follows:

$$D_0 \left(\sum_{i=0}^n a_i x^i \right) = \sum_{i=0}^n a'_i x^i; \quad D_1 \left(\sum_{i=0}^n a_i x^i \right) = \sum_{i=1}^n i a_i x^{i-1}.$$

It is not difficult to prove the following proposition:

Proposition 4. D_0 and D_1 are derivations in the ring $F[x]$.

The following theorem shows how, given an algebraic extension, it is possible to extend the derivation on the base field in a unique manner.

Theorem 3. Let $(F, ')$ be a differential field, let $E \mid F$ be an algebraic extension. Then there exists a unique map $*$: $E \rightarrow E$ such that $(E, *)$ is a differential extension of $(F, ')$.

Proof. We prove only uniqueness (the proof of existence can be found in [1]). Suppose that there exists a derivation $*$ in E that extends the one on F . Let $b \in E$ and let $f \in F[x]$ be its minimal polynomial over F (it exists since b is algebraic over F). Then, using the properties of the derivation, one has that

$$0 = (f(b))^* = \left(\sum_{i=0}^n a_i b^i \right)^* = \sum_{i=0}^n a'_i b^i + b^* \sum_{i=1}^n a_i b^{i-1} i = (D_0 f)(b) + b^* (D_1 f)(b),$$

from which one obtains that

$$b^* = -\frac{(D_0 f)(b)}{(D_1 f)(b)}.$$

(Note that, since f is the minimal polynomial of b over F , then $D_1 f$ is a polynomial of degree less than f , hence it cannot be zero.) Thus, if there exists a derivation extending $'$ on $F[b]$, it is unique. Now, since $E \mid F$ is algebraic, for every $b \in E$ one has that the derivation extends to $F[b]$ in a unique manner. By the arbitrariness of b one obtains the thesis. $\square$

Observation. The previous theorem also provides us with a way to compute the derivative of algebraic elements in a purely abstract manner. Let us think for example of the differential field of fractions of polynomials with coefficients in $\mathbb{C}$ with the usual derivative $(\mathbb{C}(x), ')$. The element $b = \sqrt{x}$ does not belong to the field but is a root of the polynomial $P(t) = t^2 - x \in \mathbb{C}(x)[t]$. We note moreover that, since P is monic and irreducible, it is also the minimal polynomial of b over $\mathbb{C}(x)$. One immediately finds that $(D_0)(b) = -1$ and $(D_1)(b) = 2b$. Hence, by what was proved above,

$$(\sqrt{x})' = b^* = -\frac{1}{2b} = \frac{1}{2\sqrt{x}}.$$

4 Liouville's Theorem

Definition 4. Let $(F, ')$ be a differential field and E a differential extension of it. An element $t \in E$ is:

- a **logarithm** over F if there exists $b \in F^*$ such that $t' = b'/b$, and we write $t = \log(b)$;
- an **exponential** over F if there exists $b \in F$ such that $t' = tb'$ and we write $t = e^b$;
- **elementary** over F if it is algebraic or an exponential or a logarithm.

Definition 5. E is an **elementary extension** of F if there exist $t_1, \dots, t_n$ in E such that $E = F(t_1, \dots, t_n)$ and one has that t_i is elementary over $F(t_1, t_2, \dots, t_{i-1})$. We will say moreover that $f \in F$ is **elementarily integrable** over F if there exists an elementary extension $E \mid F$ and $g \in E$ such that $g' = f$. Finally, an **elementary function** is any element of an elementary extension of $(\mathbb{C}(x), d/dx)$.

For example $e^{(e^x)}$ is elementary (two exponential extensions of $\mathbb{C}(x)$), as is $\sqrt{\log(x)}$ (a logarithmic extension followed by an algebraic one).

We can now define precisely the problem of integration in finite terms: given a differential field F and $f \in F$, the problem consists in deciding in a finite number of steps whether f is elementarily integrable over F and in computing its antiderivative, if it exists.

Observation. Consider $F = \mathbb{C}(x, t_1, t_2)$ where x, t_1, t_2 are indeterminates over $\mathbb{C}$, with the derivation D such that $Dx = 1$, $Dt_1 = t_1$, $Dt_2 = t_1/x$ (in practice $t_1 = e^x$ and $t_2 = \text{Ei}(x)$ with $\text{Ei}(x) := \int e^x/x dx$, which we will prove later not to be elementarily integrable). Then

$$\int \frac{t_1 t_2}{x} dx = \frac{t_2^2}{2} \in F,$$

and hence the element $\frac{t_1 t_2}{x}$ (which morally is $\frac{e^x \text{Ei}(x)}{x}$) is elementarily integrable over F (after all, its integral lies in F , which is certainly an elementary extension of F) but is not an elementary function, since it is not an element of an elementary extension of $(\mathbb{C}(x), d/dx)$.

Observation. The elementary functions of Definition 5 include all the usual elementary functions of analysis, since the trigonometric functions and their inverses can be rewritten in terms of complex exponentials and logarithms (thanks to the celebrated formula of Euler $e^{fi} = \cos(f) + i \sin(f)$).

Lemma 1. Let $(F, ')$ be a differential field. Let $F(t)$ be a differential extension of F with t transcendental over F such that $C_F = C_{F(t)}$.

1. If $t' \in F$, then for every polynomial $f(t) \in F[t]$ of degree $n > 0$, $(f(t))'$ is a polynomial in $F[t]$ of degree n or $n - 1$.

2. If $t'/t \in F$, then for every $a \in F$ one has that $(at^n)' = ht^n$ for some $h \in F$, and for every polynomial $f(t) \in F[t]$ one has that $(f(t))'$ is a polynomial in $F[t]$ of the same degree as $f(t)$.

Proof. 1. Let $t' = b \in F$ and $n = \deg f(t)$. Then we can write $f(t) = a_nt^n + a_{n-1}t^{n-1} + \dots + a_0$ with $a_i \in F$ and $a_n \neq 0$. Then

$$f'(t) = a'_nt^n + (a_nb n + a'_{n-1})t^{n-1} + \dots + (a_{n-i}b(n-i) + a'_{n-i-1})t^{n-i} + \dots + a'_1.$$

Hence the derivative of $f(t)$ continues to be a polynomial in $F[t]$ of degree n if $a'_n \neq 0$. Suppose now that $a'_n = 0$ and suppose by contradiction that $(a_nb n + a'_{n-1}) = 0$; one would then have that $a_nb n + a_{n-1} = (na_nt + a_{n-1})' = 0$, and hence $na_nt + a_{n-1}$ would be a constant, that is, an element of $C_{F(t)}$. But since we have assumed $C_F = C_{F(t)}$, one would have that $na_nt + a_{n-1} \in F$, that is, $t \in F$, which is absurd since t is transcendental over F ; hence if a_n is a constant, then the degree of $(f(t))'$ is $n - 1$.

2. Suppose now that $t'/t = b \in F$. Let $0 \neq a \in F$. Then

$$(at^n)' = a't^n + ant^{n-1}t' = (a' + abn)t^n.$$

Now if $a' + abn = 0$, then $(at^n)' = 0$, and hence at^n would be a constant and therefore an element of F , contradicting the hypothesis of transcendence of t over F . Hence $a' + abn \neq 0$. Therefore one has that if $f(t) \in F[t]$, then $(f(t))'$ is a polynomial of the same degree as $f(t)$. $\square$

We now prove the central result of the thesis.

Theorem 4 (Liouville). *Let $(F, ')$ be a differential field and let $f \in F$. The following two assertions are equivalent:*

- (i) *There exists an elementary extension $E \mid F$ such that $C_E = C_F$ and $g \in E$ with $g' = f$.*
- (ii) *There exist $v \in F$, $u_1, \dots, u_n \in F^*$ and $c_1, \dots, c_n \in C_F$ such that*

$$f = v' + \sum_{i=1}^n c_i \frac{u_i'}{u_i}.$$

Proof. (ii) $\Rightarrow$ (i). It suffices to set

$$g = v + \sum_{i=1}^N c_i \log(u_i).$$

In this way $g' = f$.

(i) $\Rightarrow$ (ii). Since the extension $E \mid F$ is elementary, there exist $t_1, t_2, \dots, t_m \in F$ such that $E = F(t_1, \dots, t_m)$ and, for every $i = 1, \dots, m$, t_i is an algebraic, or exponential, or logarithmic element over $F(t_1, \dots, t_{i-1})$. Moreover, since E has the same subfield of constants as F , then every differential field L such that $E \mid L \mid F$ will have the same subfield of constants. We prove the theorem by induction on m . For $m = 0$ one has that $E = F$ and hence the theorem is trivially proved: with $v = y$. We assume the theorem true for every elementary field extension $F(t_1, \dots, t_m)$ and we consider the chain of extensions $F(t_1, \dots, t_m) \mid F(t_1) \mid F$. By the inductive hypothesis there exist $c_1, \dots, c_n \in C_F$, $u_1, \dots, u_n \in F(t_1)$, $v \in F(t_1)$ such that $f = v' + \sum_{i=1}^n c_i \frac{u_i'}{u_i}$. We must prove that such an expression for f exists with $u_1, \dots, u_n \in F$, $v \in F$ (possibly for a different n). Let $t = t_1$; we divide the proof into three cases: the one in which t is algebraic, the one in which t is a logarithm, and finally the one in which t is an exponential.

The algebraic case. Since t is algebraic over F , there exist $U_1, \dots, U_n, V \in F[x]$ such that $U_1(t) = u_1, \dots, U_n(t) = u_n$, $V(t) = v$. We now consider the minimal polynomial of t over F ,

$p_t = \min_F(b)$, and the distinct conjugates of t , namely $\tau_1 (= t), \tau_2, \dots, \tau_s$ in K , the splitting field of p_t over F . Since $K \mid F$ is algebraic, by Theorem 3 the derivation on F can be extended in a unique manner on F . By hypothesis one then has that

$$f = (V(\tau_1))' + \sum_{i=1}^n c_i \frac{(U_i(\tau_1))'}{U_i(\tau_1)}. \quad (1)$$

Since the polynomial p_t is separable and irreducible over F , then by Proposition 1 the subgroup of S_n isomorphic to the Galois group $\text{Gal}(K \mid F)$ is transitive, and hence for every τ_j conjugate of τ_1 there exists $\sigma \in \text{Gal}(K \mid F)$ such that $\sigma(\tau_1) = \tau_j$. Applying σ to both members of equation (1) one has:

$$\begin{aligned} \sigma(f) &= f = \sigma \left((V(\tau_1))' + \sum_{i=1}^n c_i \frac{(U_i(\tau_1))'}{U_i(\tau_1)} \right) \\ &= \sigma((V(\tau_1))') + \sum_{i=1}^n \sigma(c_i) \frac{\sigma(U_i(\tau_1))'}{\sigma(U_i(\tau_1))} \\ &= \sigma(V(\tau_1))' + \sum_{i=1}^n c_i \frac{\sigma((U_i(\tau_1))')}{U_i(\tau_j)}. \end{aligned}$$

With the notation of Theorem 3 we write more explicitly what $\sigma(V(\tau_1))'$ is:

$$\begin{aligned} \sigma(V(\tau_1))' &= \sigma((D_0V)(\tau_1) + (D_1V)(\tau_1)\tau_1') \\ &= D_0V(\sigma(\tau_1)) + D_1V(\sigma(\tau_1)) \left(-\frac{(D_0V)(p_t)}{(D_1V)(\sigma(p_t))} \right) \\ &= D_0V(\tau_j) + D_1V(\tau_j) \left(-\frac{D_0V(p_t)}{D_1V(p_t)} \right) = (V(\tau_j))'. \end{aligned}$$

In an analogous way one has that $\sigma(U_i(\tau_1))' = (U_i(\tau_j))'$. One thus has that

$$f = (V(\tau_j))' + \sum_{i=1}^n c_i \frac{(U_i(\tau_j))'}{U_i(\tau_j)}, \quad \forall j \in \{1, \dots, s\}.$$

Now summing over j on both members and dividing by s , one obtains

$$\begin{aligned} f &= \frac{(V(\tau_1))'}{s} + \dots + \frac{(V(\tau_s))'}{s} + \sum_{i=1}^n \frac{c_i}{s} \left(\frac{(U_i(\tau_1))'}{U_i(\tau_1)} + \dots + \frac{(U_i(\tau_s))'}{U_i(\tau_s)} \right) \\ &= \left(\frac{V(\tau_1) + \dots + V(\tau_s)}{s} \right)' + \sum_{i=1}^n \frac{c_i}{s} \frac{(U_i(\tau_1) \cdots U_i(\tau_s))'}{U_i(\tau_1) \cdots U_i(\tau_s)}, \end{aligned}$$

where in the second passage point 5 of Proposition 3 was used. Now we note that $U_i(x_1) \cdots U_i(x_s)$ and $V(x_1) + \dots + V(x_s)$ are symmetric polynomials in $F[x_1, \dots, x_s]$, and since $\tau_1, \dots, \tau_s$ are the roots of the minimal polynomial of t over F , then by Theorem 1 one has that $U_i(\tau_1) \cdots U_i(\tau_s), V(\tau_1) + \dots + V(\tau_s) \in F$, and hence one has the thesis.

Before passing to the logarithm and exponential cases, it is convenient to make a preliminary observation. Suppose that t is transcendental over F . Then by the inductive hypothesis we know that there exist $u_1(t), \dots, u_n(t), v(t) \in F(t)$ and $c_1, \dots, c_n \in C_F$ such that

$$f = (v(t))' + \sum_{i=1}^n c_i \frac{(u_i(t))'}{u_i(t)}. \quad (2)$$

Now, since $u_i(t) \in F(t)$ for every $i = 1, \dots, n$, there exist $p_i, q_i \in F[t]$ such that $u_i(t) = \frac{p_i}{q_i}$. Considering the factorisation into irreducibles of p_i and q_i one obtains

$$u_i(t) = \frac{p_i}{q_i} = \frac{p_{1i}^{\alpha_{1i}} \cdots p_{si}^{\alpha_{si}}}{q_{1i}^{\beta_{1i}} \cdots q_{ri}^{\beta_{ri}}} = p_{1i}^{\alpha_{1i}} \cdots p_{si}^{\alpha_{si}} q_{1i}^{-\beta_{1i}} \cdots q_{ri}^{-\beta_{ri}},$$

from which it follows, using the properties of Proposition 3, that

$$\frac{(u_i(t))'}{u_i(t)} = \sum_{j=1}^s \alpha_{ji} \frac{p'_{ji}}{p_{ji}} - \sum_{k=1}^r \beta_{ki} \frac{q'_{ki}}{q_{ki}}.$$

From this observation it follows that the summation in equation (2) can be rewritten in the form

$$\sum_{i=1}^N \tilde{c}_i \frac{\tilde{u}'_i}{\tilde{u}_i},$$

in which for every $i = 1, \dots, N$, $\tilde{c}_i \in C_F$ and $\tilde{u}_i \in F[t]$ is monic and irreducible (or directly an element of F). We now consider Theorem 2, which tells us that it is possible to decompose v into partial fractions:

$$v = T + \sum_{i=1}^n \sum_{j=1}^{k_i} \frac{S_{ij}}{P_i^j}.$$

We can therefore rewrite equation (2) in the following way:

$$f = T' + \sum_{i=1}^N \tilde{c}_i \frac{\tilde{u}'_i}{\tilde{u}_i} + \sum_{i=1}^n \sum_{j=1}^{k_i} \left(\frac{S'_{ij}}{P_i^j} - \frac{j S_{ij} P'_i}{P_i^{j+1}} \right). \quad (3)$$

The logarithm case. Suppose that $t' = a'/a$ (that is, $t = \log(a)$) for a certain $a \in F$; then $t' \in F$. By Lemma 1 one has that for every monic polynomial $p(t) \in F[t]$ of degree $n > 0$, $(p(t))'$ is a polynomial in $F[t]$ of degree $n - 1$. From this it follows that $p(t)$ does not divide $(p(t))'$. Applying this reasoning to one of the P_i in the partial fraction decomposition of v , one has that in equation (3) a term with $P_i^{k_i+1}$ appears in the denominator. Since there is no term that cancels $P_i^{k_i+1}$, then it must appear in the expression of f , but this is absurd because f does not depend on t . From this it follows that $v = T \in F[t]$, and therefore

$$f = T' + \sum_{i=1}^N \tilde{c}_i \frac{\tilde{u}'_i}{\tilde{u}_i}.$$

Now, reapplying the previous reasoning to $\tilde{u}_i$, one has that there is no term on the right that cancels $\tilde{u}_i$, and hence $\tilde{u}_i$ will appear in the expression of f . Therefore $\tilde{u}_i \in F$ for every $i = 1, \dots, N$. We have thus proved that

$$T' = f - \sum_{i=1}^N \tilde{c}_i \frac{\tilde{u}'_i}{\tilde{u}_i} \in F.$$

By Lemma 1, it follows that $T = ct + d$ with $c \in C_F$ and $d \in F$. Finally, then,

$$f = \sum_{i=1}^N \tilde{c}_i \frac{\tilde{u}'_i}{\tilde{u}_i} + c \frac{a'}{a} + d',$$

which is an expression of the desired form.

The exponential case. Suppose now that $t = e^b \in F$. Then, again by Lemma 1, one has that for every monic, irreducible polynomial $f(t) \in F[t]$ different from t , $f(t)$ does not divide $(f(t))'$. Reasoning in a manner analogous to the case above, one can conclude that $v \in F[t]$ and $\tilde{u}_i \in F$ for every i , or one of them is equal to t . One concludes that

$$v' = f - \sum_{i=1}^N \tilde{c}_i \frac{\tilde{u}_i'}{\tilde{u}_i} \in F,$$

and, again by Lemma 1, one has that $v \in F$, obtaining an expression of the desired form. $\square$

Using this theorem one can prove an analogous result in the case in which the field of constants C_F is algebraically closed.

Theorem 5. *Let $(F, ')$ be a differential field with field of constants C_F algebraically closed and let $f \in F$. If there exists an elementary extension E of F and $g \in E$ such that $g' = f$, then there exist $v \in F$, $u_1, \dots, u_n \in F^*$ and $c_1, \dots, c_n \in C_F$ such that*

$$f = v' + \sum_{i=1}^n c_i \frac{u_i'}{u_i}.$$

The following theorem, which generalises Liouville's result by removing the restriction on the subfield of constants, was proved by Risch in 1970.

Theorem 6 (Risch, 1970). *Let $(F, ')$ be a differential field, C its field of constants and $f \in F$. If there exists an elementary extension E of F and $g \in E$ such that $g' = f$, then there exist $v \in F$, $c_1, \dots, c_n \in \overline{C}$ and $u_1, \dots, u_n \in F(c_1, \dots, c_n)^*$ such that*

$$f = v' + \sum_{i=1}^n c_i \frac{u_i'}{u_i}.$$

A proof of these last two results can be found in [1].

5 Examples of functions that are not elementarily integrable

In this chapter we will examine the differential field $\mathbb{C}(x)$ with the standard derivation, which we will denote by the classical symbol $'$. Moreover, from now on we will consider functions f of a real variable with complex values. Clearly an f of this type can be decomposed as

$$f = \operatorname{Re} f + i \operatorname{Im} f.$$

The reason for plunging into the complex world is simple: in this world the trigonometric functions can be traced back to exponential ones, just as their inverses reduce to logarithms. In particular, the well-known formula of Euler gives us the following relations:

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}; \quad \cos(x) = \frac{e^{ix} + e^{-ix}}{2}; \quad \tan(x) = \frac{e^{ix} - e^{-ix}}{e^{ix} + e^{-ix}};$$

$$\arcsin(x) = \frac{\pi}{2} + i \ln(ix + \sqrt{1 - x^2}); \quad \arctan(x) = \frac{i}{2} [\ln(1 - ix) - \ln(1 + ix)].$$

Theorem 7. *Let $g \in \mathbb{C}(x)$ be a non-constant rational function. Then the function e^g is transcendental over $\mathbb{C}(x)$.*

A proof can be found for example in [4].

We now want to derive a criterion to establish in which cases $f(x)e^{g(x)}$ admits an elementary antiderivative, with $f(x)$ and $g(x)$ rational functions, $f(x) \neq 0$ and $g(x)$ non-constant. We write $e^g = t$, in this way we have $t'/t = g'$, and we imagine working in the differential field $\mathbb{C}(x, t)$, a transcendental (but still elementary) extension of $\mathbb{C}(x)$.

Theorem 8. *Let $f, g \in \mathbb{C}(x)$, $f \neq 0$, g non-constant. Then the function $h(x) = f(x)e^{g(x)}$ admits an elementary antiderivative if and only if there exists $a \in \mathbb{C}(x)$ such that $f = a' + ag'$.*

Proof. ($\Leftarrow$) Writing $e^g = t$, we have that $t' = tg'$. Suppose now that there exists $a \in \mathbb{C}(x)$ such that $f = a' + ag'$. Then

$$(at)' = a't + at' = a't + ag't = (a' + ag')t = ft,$$

and hence at is an antiderivative of h and is an elementary function (which lies in $\mathbb{C}(x, t)$, which is elementary by Definition 5).

($\Rightarrow$) Suppose that h admits an elementary antiderivative $y \in E$ with E an elementary extension of $\mathbb{C}(x, e^g)$. Then by Liouville's theorem there exist $c_1, \dots, c_n \in \mathbb{C}$, $u_1, \dots, u_n, v \in \mathbb{C}(x, e^g)$ such that

$$fe^g = v' + \sum_{i=1}^n c_i \frac{u_i'}{u_i}. \quad (4)$$

Since e^g is a non-algebraic and exponential element over $\mathbb{C}(x)$, recalling the proof of Liouville's theorem, one can rewrite equation (4) as

$$fe^g = v' + \sum_{i=1}^N d_i \frac{v_i'}{v_i},$$

with $d_i \in \mathbb{C}$, $v_i \in \mathbb{C}(x)$ (or one of them equal to e^g) and $v \in \mathbb{C}(x)[e^g]$. Setting $d = \sum_{i=1}^N d_i \frac{v_i'}{v_i} \in \mathbb{C}(x)$, one has

$$fe^g = d + v'.$$

Since the left-hand member is a polynomial in $\mathbb{C}(x)[e^g]$ of degree 1, then the right-hand member must be so too. Hence one has that $v = s_0 + s_1 e^g$ with $s_0, s_1 \in \mathbb{C}(x)$. Therefore one obtains

$$fe^g = d + (s_0 + s_1 e^g)' = d + s_0' + s_1' e^g + s_1 g' e^g,$$

from which

$$e^g(f - s_1' - s_1 g') = d + s_0'. \quad (5)$$

Since $f - s_1' - s_1 g'$ and $d + s_0'$ are rational functions while e^g is not, one has that equation (5) is satisfied if and only if

$$f - s_1' + s_1 g' = 0.$$

Setting $a := s_1$ one has the thesis. $\square$

Notable examples

Using Theorem 8 it is possible to prove that there does not exist an elementary antiderivative for some well-known functions.

$$e^{x^2}$$

Our objective is to deny the existence of a rational function $a = \frac{P}{Q}$ such that $a' + ag' = e^{x^2}$, with P and Q polynomials with complex coefficients, which we assume to be coprime. Since $g(x) = x^2$ and $f(x) = 1$, the identity that they would solve would be

$$\frac{P'Q - PQ'}{Q^2} + \frac{2xP}{Q} = 1,$$

which is equivalent to

$$P'Q - Q'P + 2xPQ = Q^2. \quad (6)$$

From this relation one notes that, since Q divides Q^2 , then Q should divide the polynomial $Q'P$. Since however Q and P are coprime, Q should divide Q' . On the other hand this would imply that $\deg(Q) \leq \deg(Q')$, and this is possible only if Q is a constant polynomial. Hence the identity (6) would become

$$P'(x) + 2xP(x) = 1.$$

Clearly, however, no polynomial can satisfy this last equality. It is not difficult to see that nothing changes if instead of e^{x^2} we take e^{-x^2} , that is, the Gaussian.

$$\frac{e^x}{x}$$

Suppose there exists a rational function $a \in \mathbb{C}(x)$ such that

$$\frac{1}{x} = a' + a. \quad (7)$$

Then, writing a as $\frac{P}{Q}$ with $P, Q \in \mathbb{C}[x]$ coprime, one can rewrite (7) as

$$\frac{1}{x} = \frac{P'Q - PQ'}{Q^2} + \frac{P}{Q} = \frac{PQ + P'Q - PQ'}{Q^2},$$

which becomes

$$1 = \frac{x(PQ + P'Q - PQ')}{Q^2} \quad \text{that is,} \quad Q^2 = x(PQ + P'Q - PQ').$$

In particular one has that Q^2 must divide the numerator, but Q^2 does not divide x inasmuch as the first is a product of second-degree polynomials and the second is a linear polynomial. Moreover $Q(x)$ divides neither Q' (again because $\deg(Q) \leq \deg(Q')$) nor P (because we assumed that P, Q are coprime). Hence, as before, one has that $a \in \mathbb{C}[x]$. Equation (7) becomes of the form

$$\frac{1}{x} = P + P'.$$

Since the right-hand member of the equation is a polynomial and the left-hand member is a rational function, one can conclude that there does not exist any $a \in \mathbb{C}[x]$ that satisfies the equality, and hence by Theorem 8 the function $\frac{e^x}{x}$ does not admit an elementary antiderivative.

$$\frac{\sin x}{x}$$

One finds

$$\int \frac{\sin x}{x} dx = \int \frac{e^{ix} - e^{-ix}}{2ix} dx = \frac{1}{2i} \left(\int \frac{e^{ix}}{x} dx - \int \frac{e^{-ix}}{x} dx \right),$$

which, by what was seen previously, is not elementarily integrable.

$$\frac{e^x}{x^n}, n \in \mathbb{N}$$

For $n > 0$ we define I_n as follows:

$$I_n = \int \frac{e^x}{x^n} dx.$$

By means of integration by parts we can write

$$I_n = \frac{1}{n-1} \left(\int \frac{e^x}{x^{n-1}} dx - \frac{e^x}{x^{n-1}} \right) = \frac{1}{n-1} \left(I_{n-1} - \frac{e^x}{x^{n-1}} \right).$$

One notes that I_n is elementarily integrable if and only if I_{n-1} is. Iterating this procedure $n-1$ times we arrive at proving that I_n is elementarily integrable if and only if I_1 is, that is, e^x/x , which we have proved not to be elementarily integrable.

$$\frac{1}{\ln^n(x)}, n \in \mathbb{N}$$

As regards this function, with a substitution one has that

$$\int \frac{dt}{\ln^n(t)} \stackrel{\ln(t)=x}{=} \int \frac{e^x}{x^n} dx.$$

And hence, by what was proved previously, $\frac{1}{\ln^n(x)}$ too is not elementarily integrable.

Elliptic Integrals

The theory of elliptic functions arises from the computation of the length of the ellipse. They can be obtained by inverting functions of the form $\int dx/\sqrt{P(x)}$ for particular polynomials P with $P \in \mathbb{R}[x]$, without multiple roots. In general, if $P \in \mathbb{R}[x]$ has degree ≥ 3 and has all simple roots, then $\int dx/\sqrt{P(x)}$ is not elementarily integrable. Since $F := \mathbb{C}(x, \sqrt{P})$ is a differential field, by Liouville's theorem it is sufficient to prove that there does not exist an identity of the form

$$\frac{1}{\sqrt{P(x)}} = v' + \sum_{i=1}^n c_i \frac{u_i'}{u_i},$$

with $c_1, \dots, c_n \in \mathbb{C}$ and $u_1, \dots, u_n, v \in F$. From the theory of compact Riemann surfaces one deduces the impossibility of such an identity. For the details one can see [3].

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